

GIMIN ROH

+1-949-400-7028 | giminr@uci.edu | website_gimin
4060, 653 E Peltason Dr, Irvine, CA 92697

RESEARCH INTEREST

- **Sustainable Transportation System:** Electrified Transportation, Shared Mobility, Automated Vehicles
- **Interaction of Transportation, City, and Energy:** Transportation-Infrastructure Planning, Energy Networks
- **Optimization & AI:** Human-Based Optimal Operations, Network Modeling & Optimization, ML, DL

EDUCATION

- **University of California, Irvine** Sep. 2025 - Present
Transportation Systems, Civil and Environmental Engineering | Doctor of Philosophy California, United States
- **Hongik University** Jan. 2019 - Feb. 2025
Urban Engineering | Bachelor of Science (GPA 4.05 / 4.5; Major GPA 4.29 / 4.5) Seoul, Korea
Leave of absence for military service & working holiday: Jan. 2021 - Feb. 2023

PUBLICATIONS

P=Refereed Journal Articles, UR=Under Review, WP=Working Papers

- [P.2] **Gimin Roh**, Eunseo Ko, and Minhee Kang. 2025. *No Safety Zone for Old Man: Developing an Explainable Safety Index for Selecting Elderly Pedestrian Safety Zone*. Safety Science, 187, 106843. <https://doi.org/10.1016/j.ssci.2025.106843>
- [P.1] **Gimin Roh**, Sujae Kim, and Sangho Choo. 2024. *Optimal Supply Calculation of Electric Vehicle Slow Chargers Considering Charging Demand Based on Driving Distance*. The Journal of The Korea Institute of Intelligent Transport Systems, 23(2), 142-156. <http://dx.doi.org/10.12815/kits.2024.23.2.142>
- [WP.2] **Gimin Roh**, Matthew Dean. 2026. *Bridging the Safety Gap in Youth E-Bike Use: A Survey-Based Study on Behavioral Patterns, Education, and Risk Awareness*
- [WP.1] **Gimin Roh**, Sujae Kim, and Sangho Choo. 2025. *Optimizing the Location and Capacity of EV Charging Stations with Fast Chargers in Two Types of Highway Rest Areas Considering Both User and Operator Perspectives*

CONFERENCE PAPERS AND PRESENTATIONS

- [CP.4] **Gimin Roh**, and Sujae Kim. 2025. *Optimizing Highway EV Charging Stations: Hierarchical Approach for Locating and Allocating Fast Chargers*. 2025 Suwon ITS AP Forum.
- [CP.3] Eunseo Ko, Gitae Park, **Gimin Roh**, and Sangho Choo. 2024. *Can public ridership be predicted by Urban Imagery and Deep Learning?: A Separable Latent Vector Approach*. Korean Society of Intelligent Transportation Systems Conference.
- [CP.2] **Gimin Roh**, Eunseo Ko, Kanglim Lim, Yejoon Kim, Minseo Cho and Changsoo Ok. 2024. *A Flow-Demand-Based Highway Hydrogen Refueling Station Location Selection Model Considering the Ecosystem of Liquid Hydrogen*. Korean Institute of Industrial Engineers Conference.
- [CP.1] **Gimin Roh**, Sujae Kim, and Sangho Choo. 2023. *Calculation of Electric Vehicle Slow Charger Demand Considering Daily Car Travel Distance Distribution*. Proceedings of the Korean Society of Transportation Conference, 59-59.

RESEARCH EXPERIENCES

Graduate Research Assistant

- **Sustainable Transportation & Evolving Energy Research Lab** | Advisor: Prof. Matthew D Dean Fall 2025 - Present
Institute of Transportation Studies, University of California, Irvine

Undergraduate Research Assistant

- **Urban Transportation & Mobility Lab** | Advisor: Prof. Sangho Choo Spring 2023 - Summer 2025
Department of Urban Planning, Hongik University
 - **Optimization of Charging Station Location and Development of Optimal Path Search Algorithm Considering the Characteristics of Eco-Friendly Vehicle** PI: Sujae Kim, Ph.D
Funded: National Research Foundation of Korea(2023.06 - 2026.05, \$150,000)
 - **Public Transportation Status Survey** PI: Gyeongjae Lee, Ph.D
Funded: Korea Transportation Safety Authority

- Sustainable Mobility as a Service (S-MaaS) Strategy Development
- Policy Analysis of Road Opening & Highway Expansion
Bugang Station - North Dajeon IC Road | Yeongdong Highway (Seochang - Ansan Section) PI: Jahun Koo, Ph.D Candidate
- **Automated Vehicle & Artificial Intelligence Lab** | Advisor: Prof. Keeyeon Hwang Jul. 2023 - Jun. 2024
Department of Urban Planning, Hongik University
- Development of an Explainable Safety Index Using Artificial Intelligence PI: Minhee Kang, Ph.D

Undergraduate Student Research Program

- Undergraduate Research Opportunities Program Advisor: Prof. Sangho Choo
Hongik University Sep. 2024 - Dec. 2024
- Graduation Research Program Advisor: Prof. Sangho Choo
Hongik University Spring 2024, Fall 2024

SELECTED HONORS AND AWARDS

\$1 ≈ 1,000 KRW

- **Graduate Dean's Recruitment Fellowship**, University of California Irvine \$5000 | Sep. 2025
- **The President's Award of the Korea Transport Institute**, Graduation Research Program \$300 | Nov. 2024
- **Undergraduate Student Conference Presentation Competition** \$200 | Oct. 2024
| 1st Place [CP.3] Korean Society of Intelligent Transportation Systems
- **Undergraduate Student 'Industrial Engineering' Project Competition** Oct. 2024
| Honorable Mention [CP.2] Korean Institute of Industrial Engineers
- **Public Diplomacy Data Utilization Competition | 1st Place** \$ 3,000 | Sep. 2024
Title: Regional Data Analysis System for Enhancing the Business Execution Capabilities of CSOs Ministry of Foreign Affairs
- **Land and Transportation Data Utilization Competition | 2nd Place** \$ 3,000 | Aug. 2024
Title: SI-Map, Seamless Navigator Utilizing Integrated Indoor-Outdoor Maps Ministry of Land, Infrastructure and Transport
- **Public Data Utilization and Analysis Idea Competition | 1st Place** \$ 2,000 | Jul. 2023
Title: Selection of Priority Improvement Areas for Senior Pedestrian Traffic Safety in Seongnam City Seongnam City
- **Tutoring Program Scholarship**, Hongik University \$ 350 | Spring 2023
- **Merit Scholarship**, Hongik University(All semesters except the first) Total: \$ 14,050 | Fall 2019 - Fall 2024

SKILLS

- Python, R, LaTeX | GIS (ArcGis pro & QGis) | TransCAD, PTV VISSIM | Google OR-tools | Adobe Illustrator, Adobe Photoshop, Adobe Premiere Pro, Autodesk, Microsoft Office

EXTRACURRICULAR ACTIVITIES AND SERVICES

Teaching Experiences

- **Basic Analysis and Processing of Urban Geodata and Networks** | Guest Instructor Sungkyul University | May 2025
- **Urban Engineering Academic Society "Dosirak"** | Mentor Hongik University | Fall 2023 - Summer 2025
- **Tutoring Program "Hongik Together"** | Tutor Hongik University | Spring 2023

Working Experiences

- **Suwon City Ballet** | Strategic Planning Assistant Manager Suwon, Korea | Jan. 2024 - Aug. 2025
- **Working Holiday in New Zealand**
 - **Pure Pac** | Cherry Packer Cromwell, New Zealand | Jan. 2023 - Feb. 2023
 - **Hortinvest Horthub** | Cherry Picker Tarras, New Zealand | Dec. 2022 - Jan. 2023
 - **Skycity Casino** | Dealer (Main game: Roulette) Auckland, New Zealand | Jul. 2022 - Dec. 2022
- **Republic of Korea Army VIII Corps** | Artillery Fire Direction Specialist Goseong, Korea | Jan. 2021 - Jul. 2022
 - Served in the army to fulfill a mandatory national military service requirement; honorably discharged as Sergeant